

# Price with a Smile: A Production-style Local Volatility Surface & Backtest

## Objective

This project implements and backtests a Local Volatility (Dupire-type) pricing and hedging framework against a Black-Scholes sticky-implied-volatility benchmark, using real multi-day market option data. The primary goal is to evaluate which model produces more stable  $\Delta$ -hedged P&L and lower out-of-sample pricing error (RMSE), thus testing whether local volatility truly “captures the smile” in a realistic production setting.

## Motivation

Modern derivatives desks require robust pricing frameworks that remain consistent across time and market conditions. While the Black-Scholes model assumes constant volatility, markets exhibit *smiles and skews* that evolve daily. The Local Volatility model, derived via the Dupire equation, adapts the volatility surface dynamically to implied prices, potentially providing better hedging stability.

This project simulates the full quant workflow a junior quant researcher would execute in production: data collection  $\rightarrow$  surface fitting  $\rightarrow$  calibration  $\rightarrow$  model differentiation  $\rightarrow$  pricing  $\rightarrow$  hedging and P&L diagnostics. We also utilized HPC facilities at ASU to substantially decrease back-testing runtime.

## Methodology

1. Market Data Preparation: Clean and align daily option quotes across strikes and maturities.
2. Forward Curve Construction: Infer forwards from interest rates and spot dynamics.
3. Implied Volatility Surface: Interpolate arbitrage-free implied volatilities across strike-maturity grids.
4. Local Volatility Extraction: Apply Dupire’s partial-derivative formula to obtain the local vol surface.
5. Pricing Engine: Price vanilla options under both Local-Volatility (finite-difference PDE) and Sticky-Volatility (Black-Scholes) models.
6.  $\Delta$ -Hedging Backtest: Re-hedge daily using model-specific deltas and measure realized P&L.
7. Diagnostics: Evaluate RMSE in price space, cumulative hedged P&L, and Greeks consistency.

## Key Results

- Arbitrage-free surfaces were successfully generated for multiple consecutive trading days.
- The Local Volatility model showed smoother  $\Delta$ -hedged P&L trajectories and slightly improved RMSE over the sticky-vol baseline in several test windows.
- The code was successfully parallelized to run each day back-test on a separate CPU node on the Sol supercomputer at ASU, thus substantially decreasing runtime from 5 hours to ~12 minutes.

## Impact & Future Work

- Demonstrates a complete front-to-back quant research prototype, from data ingestion to back-testing.
- Provides a foundation for local stochastic volatility (LSV) model extensions.
- Further tweaking of parameters needs to be done to obtain better performance over sticky-volatility Black-Scholes.